

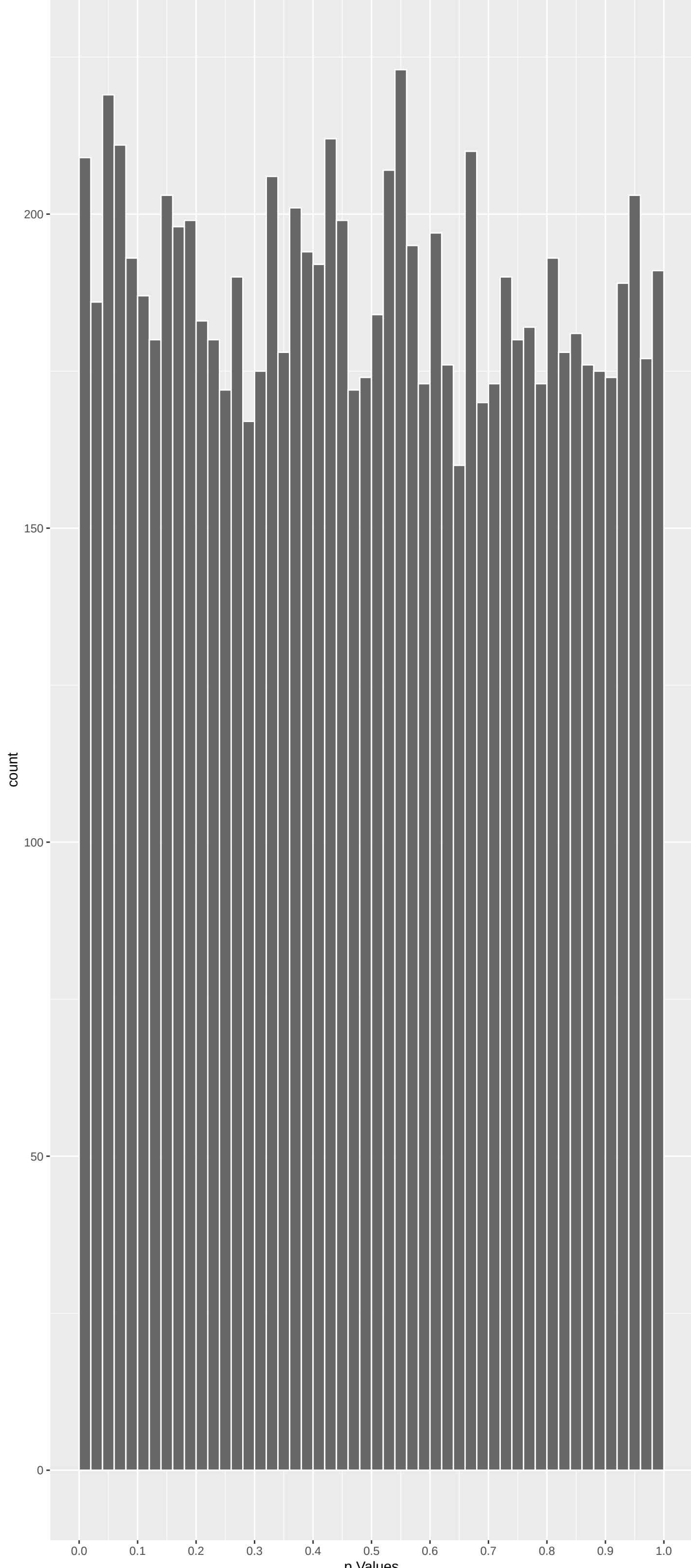
Top genes by P-value non-permulated

Gene	Rho	N	P	p.adj	qValueNoperm
DSG4	-0.5480851	174	4.969e-15	5.989e-11	5.662e-07
KRT26	-0.5572293	166	6.335e-15	5.989e-11	5.662e-07
KRT2	-0.4548186	158	1.933e-09	1.005e-05	4.752e-02
KRT76	-0.4413274	168	2.127e-09	1.005e-05	4.752e-02
TEX53	-0.4752955	132	8.475e-09	3.205e-05	1.212e-01
TAAR9	-0.4660869	134	1.383e-08	4.360e-05	1.374e-01
CPNE7	-0.4410564	136	7.691e-08	1.989e-04	3.760e-01
FAM170A	-0.4163791	151	1.052e-07	1.989e-04	3.760e-01
KRT34	-0.4746897	114	9.511e-08	1.989e-04	3.760e-01
OR2A14	-0.4292604	143	8.821e-08	1.989e-04	3.760e-01
CAPN12	-0.4034623	159	1.340e-07	2.304e-04	3.960e-01
CECR2	-0.4095942	148	2.350e-07	3.702e-04	5.833e-01
ALOXE3	-0.3760654	168	5.071e-07	7.375e-04	1.000e+00
HTT	-0.3809951	158	7.870e-07	1.063e-03	1.000e+00
TNNT2	-0.4065691	135	9.929e-07	1.252e-03	1.000e+00
OR10G9	-0.4097047	128	1.568e-06	1.853e-03	1.000e+00
GJB4	-0.3644466	158	2.507e-06	2.788e-03	1.000e+00
LTK	-0.4068875	121	3.634e-06	3.817e-03	1.000e+00
TAAR5	-0.3567611	156	4.844e-06	4.820e-03	1.000e+00
KRT75	-0.3404229	165	7.679e-06	7.260e-03	1.000e+00
CALHM3	-0.3415135	161	9.267e-06	8.275e-03	1.000e+00
MRGPRD	-0.3492985	153	9.628e-06	8.275e-03	1.000e+00
KRT83	-0.3419487	157	1.169e-05	9.212e-03	1.000e+00
LMO1	-0.5543595	55	1.127e-05	9.212e-03	1.000e+00
IAPP	-0.3812203	124	1.253e-05	9.473e-03	1.000e+00

Top genes by Q-Value non-permulated

Gene	Rho	N	P	p.adj	qValueNoperm
DSG4	-0.54808510	174	4.969e-15	5.989e-11	5.662e-07
KRT26	-0.55722934	166	6.335e-15	5.989e-11	5.662e-07
KRT2	-0.45481859	158	1.933e-09	1.005e-05	4.752e-02
KRT76	-0.44132743	168	2.127e-09	1.005e-05	4.752e-02
TEX53	-0.47529548	132	8.475e-09	3.205e-05	1.212e-01
TAAR9	-0.46608695	134	1.383e-08	4.360e-05	1.374e-01
CPNE7	-0.44105643	136	7.691e-08	1.989e-04	3.760e-01
FAM170A	-0.41637907	151	1.052e-07	1.989e-04	3.760e-01
KRT34	-0.47468966	114	9.511e-08	1.989e-04	3.760e-01
OR2A14	-0.42926039	143	8.821e-08	1.989e-04	3.760e-01
CAPN12	-0.40346228	159	1.340e-07	2.304e-04	3.960e-01
CECR2	-0.40959423	148	2.350e-07	3.702e-04	5.833e-01
A1BG	0.05979065	135	4.909e-01	9.675e-01	1.000e+00
A1CF	0.14277305	135	9.855e-02	8.090e-01	1.000e+00
A2M	0.03690044	183	6.199e-01	9.758e-01	1.000e+00
A2ML1	-0.04852633	147	5.594e-01	9.713e-01	1.000e+00
A3GALT2	-0.12359243	142	1.428e-01	8.583e-01	1.000e+00
A4GALT	-0.07077760	167	3.634e-01	9.505e-01	1.000e+00
A4GNT	-0.12184148	164	1.201e-01	8.341e-01	1.000e+00
AAAS	0.02216762	143	7.927e-01	9.905e-01	1.000e+00
AACS	0.01884007	168	8.085e-01	9.918e-01	1.000e+00
AADAC	-0.12565582	121	1.697e-01	8.714e-01	1.000e+00
AADACL2	-0.19945499	128	2.400e-02	6.115e-01	1.000e+00
AADACL3	-0.04864661	174	5.238e-01	9.711e-01	1.000e+00
AADACL4	-0.06624277	152	4.175e-01	9.559e-01	1.000e+00

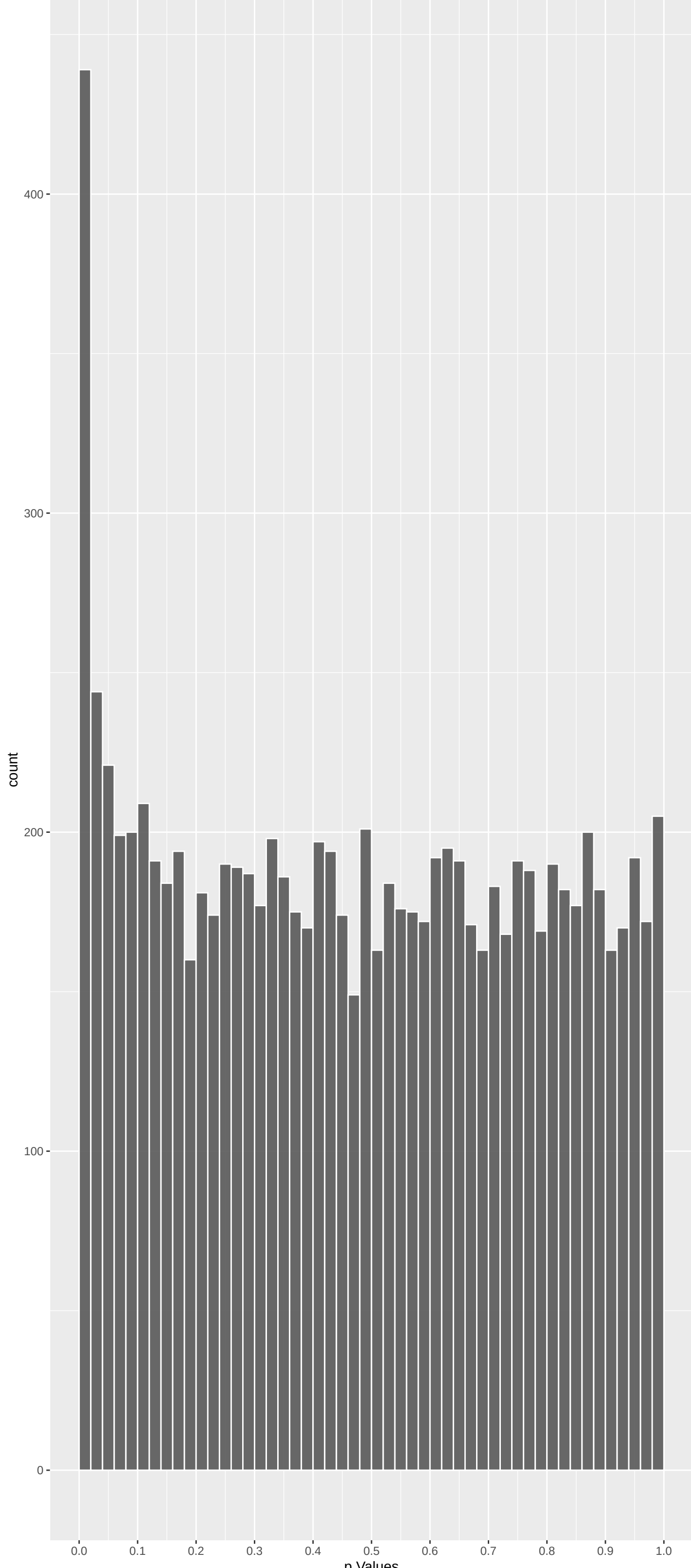
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

Gene	Rho	N	P	p.adj	qValueNoperm
ZNF628	0.3562812	141	1.450e-05	1.055e-02	1.000e+00
WNT1	0.4158184	89	5.068e-05	2.994e-02	1.000e+00
SLC26A11	0.3085501	161	6.829e-05	3.799e-02	1.000e+00
OR10S1	0.3373609	128	9.854e-05	4.333e-02	1.000e+00
SLC25A48	0.2974062	155	1.712e-04	5.885e-02	1.000e+00
FND3B	0.2874964	161	2.174e-04	6.423e-02	1.000e+00
RFPL4B	0.3365611	114	2.509e-04	7.081e-02	1.000e+00
PARP9	0.3017203	133	4.165e-04	9.488e-02	1.000e+00
C7ORF33	0.4318573	62	4.570e-04	1.005e-01	1.000e+00
ADH4	0.2764142	147	7.015e-04	1.442e-01	1.000e+00

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	N	P	p.adj	qValueNoperm
DSG4	-0.5480851	174	4.969e-15	5.989e-11	5.662e-07
KRT26	-0.5572293	166	6.335e-15	5.989e-11	5.662e-07
KRT2	-0.4548186	158	1.933e-09	1.005e-05	4.752e-02
KRT76	-0.4413274	168	2.127e-09	1.005e-05	4.752e-02
TEX53	-0.4752955	132	8.475e-09	3.205e-05	1.212e-01
TAAR9	-0.4660869	134	1.383e-08	4.360e-05	1.374e-01
CPNE7	-0.4410564	136	7.691e-08	1.989e-04	3.760e-01
FAM170A	-0.4163791	151	1.052e-07	1.989e-04	3.760e-01
KRT34	-0.4746897	114	9.511e-08	1.989e-04	3.760e-01
OR2A14	-0.4292604	143	8.821e-08	1.989e-04	3.760e-01

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Intermediate Filament Organization (GO:0	-0.21152670	65	3.774e-09	2.041e-05	KRT26:2 KRT2:3 KRT76:4 KRT34:9 KRT75:20 KRT83:24
Sensory Perception Of Smell (GO:0007608)	-0.11534385	208	1.067e-08	2.884e-05	OR2A14:8 OR10G9:16 OR51V1:62 OR8D4:66 OR2AG1:85 OR512:89
Sensory Perception Of Chemical Stimulus	-0.14629973	98	5.750e-07	1.036e-03	CALHM3:21 UGT2A1:59 OR51V1:62 OR8D4:66 OR512:89 OR5D3P:230
Regulation Of Transcription By RNA Polym	0.03399052	1948	1.400e-06	1.892e-03	PARP9:8 PHOX2A:22 CHD3:28 ZNF24:31 GADD45A:39 PROP1:45
Mitochondrial Gene Expression (GO:014005	0.12516256	102	1.283e-05	1.388e-02	GADD45GIP1:53 GF2M:288 MRPL54:328 MRPL47:359 MRPL41:384 MRPL2:513
Regulation Of DNA-templated Transcription	0.03017590	1857	2.694e-05	2.428e-02	ZNF628:1 WNT1:2 PARP9:8 PHOX2A:22 CHD3:28 ZNF24:31
Mitochondrial Translation (GO:0032543)	0.11923727	97	5.036e-05	3.026e-02	GADD45GIP1:53 GF2M:288 MRPL54:328 MRPL47:359 MRPL41:384 MRPL2:513
Positive Regulation Of DNA-templated Tra	0.03539101	1209	4.579e-05	3.026e-02	WNT1:2 PARP9:8 PHOX2A:22 CHD3:28 ZNF24:31 SOX17:66
Regulation Of Gene Expression (GO:001046	0.03725085	1085	4.430e-05	3.026e-02	ZNF628:1 RFPL4:7 PARP9:8 CHD3:28 RBM12B:41 PDCD5:42
Negative Regulation Of Gene Expression (C	0.06308747	332	8.327e-05	4.093e-02	PARP9:8 ITGB8:47 CSNK2A1:79 POU4F1:50 UGT2A1:59 TIA1:176 CAPRN1:253
Positive Regulation Of Transcription By	0.03917508	908	7.638e-05	4.093e-02	PHOX2A:22 ZNF24:31 SOX17:66 AKAP8L:106 TBX5:112 TFR2:136
Protein K6-linked Ubiquitination (GO:008	-0.37016964	9	1.203e-04	5.420e-02	UBE2D4:328 RNF4:756 RNF6:772 UBE2T:819 PRKN:1827 UBE2S:3302
Oxidative Phosphorylation (GO:0006119)	0.13965244	59	2.111e-04	8.780e-02	NDUFS2:234 ATP5F1C:316 NDUFA12:475 NDUFB10:502 NDUFB7:97 NDUFB3:906
Atrial Septum Development (GO:0003283)	0.29326003	13	2.512e-04	9.165e-02	NKX2-5:323 NOTCH2:1077 ZFPM1:1092 GATA4:1282 BMPR2:1580 MDMA:1921
Epidermis Development (GO:0008544)	-0.11556268	84	2.542e-04	9.165e-02	KRT2:3 KRT34:9 KBT83:24 GJB5:58 KRT9:116 SPRR2B:199
Outflow Tract Septum Morphogenesis (GO:0	0.20680720	24	4.598e-04	1.554e-01	NKX2-5:323 TGFBR2:827 BMP4:1269 BMPR2:1580 FGFR8:1585 ENG:2365
Cardiac Conduction System Development (G	0.20087240	25	5.089e-04	1.619e-01	TBX5:112 NKX2-5:323 SMAD5:955 NOTCH2:1077 HCN4:1096 GATA4:1282
Aorta Morphogenesis (GO:0035909)	0.23826935	17	6.679e-04	1.642e-01	TGFBR2:827 DLL4:830 JAG1:1358 RBP1:2274 ENG:2365 BMPRIA:2583
Atrial Septum Morphogenesis (GO:0060413)	0.31134335	10	6.514e-04	1.642e-01	NKX2-5:323 NOTCH2:1077 ZFPM1:1092 GATA4:1282 BMPR2:1580 TBX20:2465
Embryonic Hemopoiesis (GO:0035162)	0.27256150	13	6.675e-04	1.642e-01	FLT3LG:117 SH2B3:707 KIT:775 TGFBR2:827 ZFPM1:1092 KITLG:1312
Embryonic Organ Morphogenesis (GO:004856	0.15224434	42	6.433e-04	1.642e-01	SOX17:66 HOXC9:179 SIX1:295 HES1:544 NIPBL:561 OSR2:608
Ventricular Septum Development (GO:00032	0.16462001	36	6.330e-04	1.642e-01	TBX5:112 NKX2-5:323 HES1:544 TGFBR2:827 ZFPM1:1092 GATA4:1282
RNA Catabolic Process (GO:0006401)	0.12978303	57	7.059e-04	1.659e-01	SUPV3L1:140 LSM2:425 DSO:443 LSM5:628 RNASE6:594 SLFN14:715
Regulation Of Interleukin-2 Production (C	0.14510264	45	7.616e-04	1.716e-01	RPS3:272 CD80:284 SPTBN1:360 CD86:415 USP47:551 PRKD2:700
Negative Regulation Of Lipid Catabolic P	-0.22684988	18	8.630e-04	1.866e-01	APOC1:549 AKT1:1013 PDENB1:1422 CIDEC:1566 ALK:2011 HCAR2:2901
Proton Motive Force-Driven Mitochondrial	0.13432855	50	1.022e-03	2.125e-01	NDUFS2:234 ATP5F1C:316 NDUFA12:475 NDUFB10:502 NDUFB7:97 NDUFB3:906
Detection Of Chemical Stimulus Involved	-0.08378033	127	1.131e-03	2.264e-01	OR2A14:8 OR10G9:16 OR2AG1:85 OR13C8:107 OR2T2:126 OR10G7:133
Androgen Biosynthetic Process (GO:000670	0.33001559	8	1.227e-03	3.324e-01	SRD5A2:84 HSD17B6:356 HSD17B3:1716 MED1:3247 HSD3B1:3470 HSD3B2:3744
Detection Of Chemical Stimulus Involved	-0.08242566	129	1.246e-03	2.324e-01	OR2A14:8 OR10G9:16 OR2AG1:85 OR13C8:107 OR2T2:126 OR10G7:133
Sensory Perception Of Taste (GO:0005909)	-0.23116760	16	1.368e-03	2.466e-01	CALHM3:21 TAS1R3:75 ASIC2:321 PKD2T1:703 GNAT3:1091 CALHM1:1269
Aortic Valve Development (GO:0003176)	0.14669998	39	1.592e-03	2.474e-01	TNFRSF1A:210 NKX2-5:323 SNAIL:451 TGFBR2:827 DLL4:830 BMP4:1265
Ketone Biosynthetic Process (GO:0042181)	0.30457354	9	1.556e-03	2.474e-01	SRD5A2:84 CYP11B1:128 CYP11B2:1001 SIRT6:2992 CACNA1H:3758 TPPI:4424
Negative Regulation Of Nucleic Acid-Temp	0.04434533	444	1.445e-03	2.474e-01	CHD3:28 ZNF24:31 LMMD:148 NNOA:62 USP47:83 YEATS2:198
Regulation Of Proteolysis Involved In Pr	0.19088224	23	1.532e-03	2.474e-01	GPX1:599 PSME3:1181 PINK1:1976 RHBDF1:2316 DNAAF4:2387 TMF1:2648
Negative Regulation Of Oxidoreductase Ac	-0.28796939	10	1.614e-03	2.494e-01	DRD5:523 HDAC6:592 PRKN:1827 ETFBKM:2104 GZMA:2147 MT3:4373
Cardiac Ventricle Development (GO:000323	0.12338044	18	1.725e-03	2.521e-01	TBX5:112 HES1:544 DLL4:830 GATA4:1282 MDMA:1921 GJA5:1980
Negative Regulation Of Macromolecule Met	0.06756721	182	1.706e-03	2.521e-01	PARP9:8 ITGB8:47 POU4F1:150 FGFR13:331 CNPY2:500 SFRP1:501
Regulation Of Activated T Cell Prolifera	0.15957636	32	1.788e-03	2.544e-01	TMIGD2:99 RPS3:272 FOXPP3:551 CD274:663 RIPK3:1276 RC3H1:1430
His-Purkinje System Development (GO:0003	0.39912883	5	1.995e-03	2.587e-01	TBX5:112 NKX2-5:323 HOPX:1284 ID2:2688 IRX3:4919 NIA
Homologous Chromosome Pairing At Meiosis	-0.16297479	30	2.009e-03	2.587e-01	KASH5:579 TEX11:619 RNF212:723 HORMAD1:794 SYCP2:1441 SYCP1:1763

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_KERATINIZATION	-0.18361057	214	2.275e-20	1.478e-16	DSG4:1 KRT26:2 KRT2:3 KRT76:4 KRT34:9 KRT75:20
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	-0.18302963	128	8.844e-13	2.872e-09	DSG4:1 KRT26:2 KRT2:3 KRT76:4 KRT34:9 KRT75:20
REACTOME_OLFACTORY_SIGNALING_PATHWAY	-0.09303375	390	3.081e-10	5.001e-07	OR2A14:8 OR10G9:16 OR51T1:51 OR51V1:62 OR8D4:66 OR56A5:70
REACTOME_SENSORY_PERCEPTION	-0.07559797	605	2.428e-10	5.001e-07	OR2A14:8 OR10G9:16 CALHM3:21 OR51T1:51 OR51V1:62 OR8D4:66
KEGG_OLFACTORY_TRANSDUCTION	-0.09227588	380	7.067e-10	9.179e-07	OR2A14:8 OR10G9:16 OR51T1:51 OR51V1:62 OR8D4:66 OR56A5:70
GRAESSMANN_APOPTOSIS_BY_DOXORUBICIN_DN	0.04085370	1705	2.721e-08	2.945e-05	ARMT1:19 COL1A2:26 CCN5:30 CEMIP2:50 GABRB2:51 NONO:62
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	0.06421325	608	7.012e-08	6.505e-05	CSNK1G3:16 PDCD5:42 USP47:83 BMS1:102 GLRX:105 RTCA:114
DODD_NASOPHARYNGEAL_CARCINOMA_UP	-0.04015077	1582	1.277e-07	1.037e-04	LGALS7:29 SERPINB11:32 KRT40:46 ABCA13:54 ALDH3B2:98 KCNJ5:103
BENPORATH_NANOG_TARGETS	0.04826707	951	5.238e-07	3.780e-04	HNRNP1A:78 ANKRD1:113 TMFR2:120 INO80C:145 KLF5:147 POU4F1:150
ACEVEDO_LIVER_TUMOR_VS_NORMAL_ADJACENT_T	0.05082543	826	7.733e-07	5.022e-04	GT2F1RD2B:11 GT2F1RD2:18 HSPH1:38 PDCD5:42 H2BC21:46 PIGB:65
MARSON_BOUND_BY_FOXP3_STIMULATED	0.04530700	974	1.907e-06	9.525e-04	HSPH1:38 ABHD8:56 CTOD10:71 FRMD8:77 MYNN:97 WRAP5:133
BUYTAERT_PHOTOODYNAMIC_THERAPY_STRESS_UP	0.04979126	801	1.832e-06	9.525e-04	GOLGB1:15 ZNF24:31 KCTD10:71 GSKN2:2179 AKAP8L:106
ACEVEDO_LIVER_CANCER_UP	0.04662380	921	1.809e-06	9.525e-04	FND3C3B:6 GT2F1RD2B:11 GT2F1RD2:18 COL1A2:26 H2BC21:46 GADD45GIP1:53
BENPORATH_SOX2_TARGETS	0.05226761	706	2.417e-06	1.121e-03	H2BC21:46 HNRNP1A:78 USP49:81 MYNN:97 ANKRD1:113 SGTA:123
RODRIGUES_THYROID_CARCINOMA_ANAPLASTIC_U	0.05298271	681	2.633e-06	1.140e-03	CSNK1G3:16 LACTB:82 USP47:83 GLRX:105 ANKRD1:113 RTCA:114
MARTENS_TRETINOIN_RESPONSE_UP	0.04914749	751	4.955e-06	2.011e-03	ZNF628:1 PHOX2A:22 H2BC21:46 NONO:62 TRMT10C:63 ABT1:73
MEISSNER_BRAIN_HCP_WITH_H3K27ME3	0.08078868	258	8.097e-06	3.093e-03	PHOX2A:22 HOXC13:80 IRX6:271 SIX1:295 NKX2-5:323 WDR86:334
FLECHNER_BIOPSY_KIDNEY_TRANSPLANT_REJECT	0.05460865	535	1.636e-05	5.901e-03	ALDH5A1:44 ABCCC5:52 NONO:62 CSNK2A1:79 GLRX:105 ATP2A2:155
GABRIELY_MIR21_TARGETS	0.07456580	278	1.928e-05	6.588e-03	PARP9:8 ITGB8:47 CEMIP2:50 USP47:83 KXR3:138 KLF5:147
WONG_ADULT_TISSUE_STEM_MODULE	0.04715381	700	2.272e-05	7.377e-03	ABHD8:56 COL4A2:86 IL11RA:133 ANTXR2:152 ATP2A2:155 CEBPD:156
PID_MYC_REPRESS_PATHWAY	0.15761979	60	2.420e-05	7.482e-03	COL1A2:26 GADD45A:39 CEBPD:156 DNMT3A:300 SFRP1:501 TMEM126A:653
WP_GPCRS_CLASS_A_RHODOPSINLIKE	-0.07615207	253	3.098e-05	9.145e-03	MTNR1A:53 OR2AG1:85 ADORA2B:90 HTR4:91 OPN1MW:132 OPR1:213
REACTOME_SIGNALING_BY_INTERLEUKINS	0.05617877	463	3.582e-05	9.305e-03	COL1A2:26 GRB2:94 IL11RA:133 CEBPD:156 PTPN12:158 NANOG:196
GEORGES_TARGETS_OF_MIR192_AND_MIR215	0.04216931	843	3.470e-05	9.305e-03	FND3C3B:6 ALDH5A1:44 CEMIP2:50 NOLL:92 PPM1A:115 PLOD1:185
MITSIADES_RESPONSE_TO_APLUDIN_UP	0.05930365	415	3.532e-05	9.305e-03	GADD45A:39 AKAP8L:106 TSPYL2:111 ASCC1:175 POLR2A:178 TTC17:227
NIKOLSKY_BREAST_CANCER_TP22_AMPLICON	-0.19206421	37	5.283e-05	1.183e-02	CYP2W1:39 GNAI2:202 MICALL2:361 TMEM184A:859 FAM20C:921 SNX8:1318
LEE_LIVER_CANCER_MYC_E2F1_UP	0.15949490	54	5.038e-05	1.183e-02	COL1A2:26 IGFBP1:202 ANXA5:332 GPC3:630 PLSCR1:745 SLC1A4:864
HOLLERN_SQUAMOUS_BREAST_TUMOR	-0.09007686	170	5.144e-05	1.183e-02	KRT34:9 GJB4:17 KRT75:20 KRT83:24 KRT72:31 SERPINB11:32
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.04486473	713	4.774e-05	1.183e-02	KPNAT:25 COL1A2:26 GRB2:94 FLT3LG:117 IL11RA:133 CEBPD:156
GRAESSMANN_RESPONSE_TO_MC_AND_DOXORUBICI	0.04357760	740	5.803e-05	1.256e-02	CCN5:30 ZBTB12:68 CSNK2A1:79 UZAF1A:91 MYNN:97 CEP41:135
LEE_LIVER_CANCER_E2F1_UP	0.14665686	61	7.464e-05	1.563e-02	IGFBP1:202 ANXA5:332 RHOB:519 JCAD:522 GPC3:630 COL5A2:742
BIOCARTA_SCRPPT_PATHWAY	0.34314552	11	8.110e-05	1.646e-02	CDC25C:64 GRB2:94 CDC25A:623 CCNB1:1041 CSK:1139 CDC25B:1592
MOOTHA_MITOCHONDRIA	0.05490777	438	8.436e-05	1.660e-02	MTCH1:32 SLC25A3:87 CYP11B1:128 SUPV3L1:140 SLC30A9:171 FPGS:191
NABA_SECRETED_FACTORS	0.06190193	339	9.210e-05	1.759e-02	WNT1:2 CCL8:20 HCF2C:95 FLT3LG:117 LTB:165 S100A2:193
MARTENS_TRETINOIN_RESPONSE_UP	-0.04099227	790	9.535e-05	1.769e-02	KRT2:3 KRT34:9 GJB4:17 LMO1:23 KRT83:24 TREML4:35
REACTOME_MAPK_FAMILY_SIGNALING_CASCADES	0.06349183	319	9.899e-05	1.786e-02	GRB2:94 SHOC2:98 FLT3LG:117 SNRPB:125 VCL:242 PSME4:259
BENPORATH_MYC_MAX_TARGETS	0.04178303	749	1.056e-04	1.853e-02	H2BC21:46 CLCN2:60 ABT1:73 HNRNP1A:78 HCF2C:95 GNLI:201
GUILLAUMOND_KLF3_TARGETS_UP	0.16059470	48	1.186e-04	2.019e-02	MT1X:423 STOM:456 SYT11:546 ALDH18A1:961 SAA2:1081 CCKBR:1520
REACTOME_AMINE_LIGAND_BINDING_RECEPTORS	-0.17348509	41	1.213e-04	2.019e-02	TAAR9:6 TAAR5:19 HTR4:91 HTR1D:249 GPR143:327 HTR1B:492
REACTOME_MITOCHONDRIAL_TRANSLATION	0.11512688	93	1.251e-04	2.031e-02	GADD45GIP1:53 GF2M:288 MRPL54:328 MRPL47:359 MRPL41:384 MRPL2:513

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Diabetic Nephropathy	0.05651220	517	1.244e-05	4.585e-02	COL1A2:26 TSPYL2:111 ANKRD1:113 HNF4A:263 CD59:282 UTSR2:352
Generalized hypotonia	0.05141544	603	1.867e-05	4.585e-02	COL1A2:26 ALDH5A1:44 DHTKD1:61 NONO:62 TRMT10C:63 CSNK2A1:79
Malignant neoplasm of prostate	0.02715585	2864	6.445e-06	4.585e-02	WNT1:2 DIRAS3:13 CHD3:28 GADD45A:39 GADD45GIP1:53 CDC25C:64
Prostate carcinoma	0.02620214	2757	1.859e-05	4.585e-02	WNT1:2 DIRAS3:13 GADD45A:39 CEMIP2:50 GADD45GIP1:53 CDC25C:64
Acquired scoliosis	0.06044933	398	3.808e-05	7.480e-02	WNT1:2 COL1A2:26 ABCC5:52 NONO:62 TBX5:112 PLOD1:185
Joint laxity	0.12087824	90	7.502e-05	1.228e-01	COL1A2:26 NONO:62 PLOD1:185 RYR1:505 PAX2:555 HNRNP2H2:720
Curvature of spine	0.05779455	389	9.844e-05	1.268e-01	WNT1:2 COL1A2:26 ABCC5:52 NONO:62 TBX5:112 PLOD1:185
Pulmonary Stenosis	0.14042933	64	1.033e-04	1.268e-01	SHOC2:98 G6PC3:143 CLIP2:277 GDF1:294 GTF2I:486 GPC3:630
Thin lips	0.13794006	65	1.213e-04	1.323e-01	GDF1:294 NKX2-5:323 HDAC8:518 NIPBL:561 NOTCH2:1077 ALG1:1093
Leukemia, T-Cell	0.06128492	324	1.583e-04	1.555e-01	HNRNP1A:78 DNMT3A:300 TYR3:315 MAGI1:409 SNAP25:419 LYL1:538
Leukemia, Myelocytic, Acute	0.02858848	1511	2.725e-04	2.230e-01	WNT1:2 ABCC5:52 GADD45GIP1:53 CDC25C:64 SOX17:66 HOXC13:80
Mitochondrial Diseases	0.05599616	363	2.639e-04	2.230e-01	DHTKD1:61 TRMT10C:63 LACTB:82 COX4I2:89 PDSS1:144 PDSS2:154
Hyperreflexia	0.06111927	293	3.350e-04	2.301e-01	SPTBN2:125 NDUFS2:234 RUBCN:273 CLIP2:277 PEX16:280 PYCR2:339
Osteosarcoma	0.03581120	897	3.222e-04	2.301e-01	WNT1:2 DIRAS3:13 COL1A2:26 GADD45A:39 PDCD5:42 APOD7:4
Renal Cell Dysplasia	0.29797520	12	3.515e-04	2.301e-01	PAX2:555 CITED1:61 ESR1:638 NOTCH2:1077 BMP4:1269 JAG1:1358
Down-sloping shoulders	0.20241100	24	5.991e-04	3.269e-01	CLIP2:277 GTF2I:486 MLXIPL:755 ELN:1288 PAX1:1361 LIMK1:1945
Embryonal Carcinoma	0.07627779	171	5.938e-04	3.269e-01	SOX17:66 CREG1:130 NANOG:196 VCL:242 MAP3K14:296 PDXL:362
Glioma	0.02456327	1903	5.489e-04	3.269e-01	WNT1:2 DIRAS3:13 CCL8:20 GADD45A:39 PDCD5:42 ITGB8:47
Congenital Abnormality	0.04198249	571	6.619e-04	3.287e-01	PHOX2A:22 PROPL:45 SOX17:66 TSPYL2:111 TBX5:112 ARFGEF2:121
Mammary Neoplasms	0.02287900	2173	6.694e-04	3.287e-01	WNT1:2 DIRAS3:13 COL1A2:26 HSPH1:38 GADD45A:39 LIMD1:48
Carcinogenesis	0.01842910	3580	9.445e-04	3.944e-01	WNT1:2 FND3C3B:6 COL1A2:26 CHD3:28 ZNF24:31 GADD45A:39
Effin facies	0.30173030	10	9.531e-04	3.944e-01	CLIP2:277 GTF2I:486 ELN:1288 INSR:1824 LIMK1:1945 GTF2IRD1:2247
Fibrosarcoma	0.05768282	283	7.500e-04	3.944e-01	TBX5:112 CREG1:130 ASCC1:175 TNFRSF1A:210 SNAI1:45 FOPX3:551
Osteopenia	0.05420909	316	9.637e-04	3.944e-01	WNT1:2 COL1A2:26 PROPL:45 COX4I2:89 ANTXR2:152 LRP4:181
Phonophobia	0.28535035	11	1.049e-04	4.121e-01	CLIP2:277 GTF2I:486 MLXIPL:755 ELN:1288 EP300:1767 LIMK1:1945
COLORBLINDNESS, PARTIAL, DEUTAN SERIES	-0.16241756	32	1.478e-03	4.185e-01	OPN1MW:132 OPN1MW3:189 OPN1MW2:201 COX6A:809 HLA-DPB1:905 NOS1:1086
Body Height	0.05616316	281	1.242e-03	4.185e-01	FND3C3B:6 UTAI9:40 ZNF341:127 POLR2A:178 MICA:259 DNMT3A:300
Bunion	0.16444733	31	1.534e-03	4.185e-01	NONO:62 CLIP2:277 GTF2I:486 COL5A2:742 MLXIPL:755 COL1A1:921
Ground-glass opacification on pulmonary	-0.25426871	13	1.502e-03	4.185e-01	SFTPA1:1281 SFTPA2:1454 SFTPC:1631 MUC5B:2714 FAM13A:3018 STNL1:4164
Hallux Valgus	0.16893020	30	1.366e-03	4.185e-01	NONO:62 CLIP2:277 GTF2I:486 COL5A2:742 MLXIPL:755 COL1A1:921
Honeycomb lung	-0.25426871	13	1.502e-03	4.185e-01	SFTPA1:1281 SFTPA2:1454 SFTPC:1631 MUC5B:2714 FAM13A:3018 STNL1:4164
Hyperactive behavior	0.03690239	665	1.287e-03	4.185e-01	ALDH5A1:44 GRB2:94 SHOC2:98 C1QL1:183 SNRNP217 PRODH:219
Leukemogenesis	0.03686510	659	1.358e-03	4.185e-01	GADD45A:39 HNRNP1A:78 CXCR2:88 FLT3L:3117 DDR1:129 LTB1:65
Liver carcinoma	0.01873934	3081	1.418e-03	4.185e-01	WNT1:2 FND3C3B:6 PARP9:8 ADH4:10 DIRAS3:13 COL1A2:26
Proteinuria	0.07190904	173	1.128e-03	4.185e-01	PDSS2:154 TREX1:215 PRODH:219 HNF4A:263 CLIP2:277 ACKR1:478
Reticular pattern on pulmonary HRCT	-0.25426871	13	1.502e-03	4.185e-01	SFTPA1:1281 SFTPA2:1454 SFTPC:1631 MUC5B:2714 FAM13A:3018 STNL1:4164
Mental and motor retardation	0.03030300	971	1.590e-03	4.219e-01	WNT1:2 ALDH5A1:44 PROPL:45 DHTKD1:61 CSNK2A1:79 COLA2:86
Bicuspid aortic valve	0.12582501	49	2.321e-03	4.286e-01	PLOD1:185 NKX2-5:323 MLXIPL:755 TGFBR2:827 ACTA2:1209 ELN:1288
Bowel diverticula	0.35882531	6	2.355e-03	4.286e-01	COL5A2:742 COL1A1:923 ALDH18A1:961 ELN:1288 FBN1:592 COL5A1:9821
Bowel diverticulosis	0.35882531	6	2.355e-03	4.286e-01	COL5A2:742 COL1A1:923 ALDH18A1:961 ELN:1288 FBN1:592 COL5A1:9821